

The $\mathfrak{A}^\dagger \bar{\mathfrak{C}}^\dagger \times \mathfrak{B}_2$ row

Transverse divergences, the p^+ integral, and whether the $+$ prescription may be used

$$S \equiv p^+ + k^+, \quad A \equiv (x - z)^2, \quad B \equiv (x - w)^2, \quad C \equiv (z - w)^2, \quad \beta \equiv \frac{k^+ B}{A}, \quad \kappa \equiv \frac{k \cdot (y' - z)}{k^+}$$

Verdict. The prefactor, the $\mathfrak{C}$ delta, the phase split and the tensor contraction are all **correct except one term**: the second half of your $\frac{1}{p^+}$ bracket carries the wrong vector (§2). On divergences: **no ultraviolet divergence** anywhere — all 26 collapsing regions converge (§3) — and, unlike the previous rows, **no infrared one either**: the z integral converges as it stands, because the phase $e^{-ik \frac{p^+}{k^+} (y' - z)}$ damps it (§4). The p^+ integral is clean, six exponential–integral masters, no pole on the contour (§5). **But the answer to your last question is no.** The $+$ prescription is precisely what must *not* be applied naively here: the $p^+ \rightarrow 0$ subtraction deletes that phase and **creates** an infrared divergence the row does not have (§6).

1. Prefactor, delta and phase — all exact

ingredient	value
phases	$(-i)_{\mathfrak{A}^\dagger} (-1)_{\bar{\mathfrak{C}}^\dagger} (+i)_{\mathfrak{B}_2} = -1 \rightarrow$ the overall minus $\checkmark$
numeric	$\frac{4}{(2\pi)^3} \frac{1}{\sqrt{2\pi}} \frac{1}{2\pi\sqrt{2}} \frac{1}{4\pi^2} = \frac{1}{(2\pi)^3} \frac{1}{4\pi^4} \checkmark$
momentum	$\frac{1}{\sqrt{S}} \cdot \frac{1}{\sqrt{S}} \frac{\sqrt{p^+ k^+}}{S} \cdot \sqrt{p^+ k^+} = \frac{p^+ k^+}{S^2}$
after the delta	kernel $\frac{k^+}{S}$, Jacobian $\left(\frac{S}{k^+}\right)^2 \rightarrow$ net $\frac{p^+}{S} \checkmark$

The $\mathfrak{C}$ delta $\delta^{(2)}[(y' - w') + \frac{p^+}{S}(w' - z)]$ gives $w' = \frac{Sy' - p^+ z}{k^+}$ and $w' - z = \frac{S}{k^+}(y' - z)$, hence

$$w' - w = (y' - w) + \frac{p^+}{k^+}(y' - z) \implies e^{-ik \cdot (w' - w)} = e^{-ik \cdot (y' - w)} e^{-ik \cdot \frac{p^+}{k^+} (y' - z)} \quad \checkmark$$

exactly your two–factor phase. **Remember that second factor** — it is the whole story of §4 and §6.

2. The contraction — one term is wrong

With $M^{ij} \equiv \frac{\delta^{ij}(A-B)}{2SC} + \frac{\mathcal{K}^{ij}}{k^+} + \frac{\mathcal{P}^{ij}}{p^+}$,

$$\mathcal{K}^{ij} = \frac{(x - z)^j (z - w)^i}{C} - \frac{(x - z)^j (x - w)^i}{2B}, \quad \mathcal{P}^{ij} = \frac{(x - w)^i (z - w)^j}{C} + \frac{(x - w)^i (x - z)^j}{2A},$$

contracting the $\mathfrak{C}$ bracket $[\delta_{k'm} \delta_{ji} - \frac{S}{p^+} \delta_{jm} \delta_{k'i} - \frac{S}{k^+} \delta_{im} \delta_{k'j}]$ over i, j and multiplying by the $\frac{p^+}{S}$ of §1 gives

$$\frac{p^+}{S} \delta_{k'm} \text{tr } M - M^{k'm} - \frac{p^+}{k^+} M^{mk'}.$$

Expanded, this is your six–term bracket. Five terms match exactly, including the two $-\frac{(A-B)}{2C}$ pieces inside your $\frac{p^+ \delta_{k'm}}{k^+ S}$ and $\frac{\delta_{k'm}}{S}$ brackets, which come from the δ^{ij} part of M feeding into the $-\frac{S}{p^+}$ and $-\frac{S}{k^+}$ structures of $\mathfrak{C}$. The exception:

$$\begin{aligned} \text{you wrote} & \quad -\frac{1}{p^+} \left[\frac{(x - w)^{k'} (z - w)^m}{(z - w)^2} + \frac{(x - z)^m (z - w)^{k'}}{2(x - z)^2} \right] \\ \text{it should be} & \quad -\frac{1}{p^+} \left[\frac{(x - w)^{k'} (z - w)^m}{(z - w)^2} + \frac{(x - z)^m (x - w)^{k'}}{2(x - z)^2} \right] \end{aligned}$$

because this term is $-\frac{\mathcal{P}^{k'm}}{p^+}$, and **both** halves of $\mathcal{P}^{ij} = (x - w)^i [\frac{(z - w)^j}{C} + \frac{(x - z)^j}{2A}]$ carry the same $(x - w)^i$. The error is exactly

$$\Delta = +\frac{1}{p^+} \frac{(x - z)^{k'} (x - z)^m}{2(x - z)^2}, \quad \|\Delta\| = \frac{1}{2p^+} \text{ (configuration independent),}$$

which is why the measured discrepancy was the same number, $1.3514 = \frac{1}{2 \times 0.37}$, at every test configuration. With the fix the contraction matches to 3×10^{-16} .

2.1 Two smaller fixes

- **The measure.** You write $\int_{w,w',y',z,y,x}$. But w' has been eaten by the delta, y does not appear in the integrand, and x' is **missing** (it is in both $\frac{(x'-y')k^+}{(x'-y')^2}$ and $\rho^{e'}(x')$). It should be $\int_{x,x',y',z,w}$.
- **Transposes.** The bra bracket is the one from the two- $\mathfrak{C}$ row and carries the same $U^{d'b}(z) / U^{bd'}(z)$ and $U^{e'c'}(y') / U^{c'e'}(x')$ mismatch. The ket bracket $[f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)]$ is right, with f correctly *not* factored out (the two carry f^{cbd} and f^{acd}).

3. No ultraviolet divergence

All 26 subsets of $\{x, x', y', z, w\}$ collapsed at scale ρ against $\rho^{2(n-1)} \frac{d\rho}{\rho}$: **every one converges**. The region you had in mind, $x, z, w \rightarrow$ a point where $D = p^+ A + k^+ B$ vanishes like ρ^2 , gives integrand ρ^{-2} against a three-point measure $\rho^3 d\rho$.

Why the loop has no transverse UV. The loop gluon carries p^+ and is emitted at z in the ket ($\mathfrak{B}_2$) and absorbed at z in the bra — the $\mathfrak{C}$ delta puts the bra's emission point w' at $\frac{Sy-p^+z}{k^+}$, so the loop closes at **the same transverse point on both sides**. There is no separation to shrink, hence no short-distance singularity. The loop's large-momentum behaviour is carried entirely by the p^+ integration instead (§5), where the $h = p^+$ master grows like $\vee$.

4. And no infrared one either — the phase protects z

variable	$R = 10^2$	10^3	10^4	verdict
$ x $	0.287047	0.287065	0.287066	log divergent — but x is a ρ position, cut by the source
$ x' $			noise	convergent
$ y' $	6.2e-2	1.0e-2	1.5e-3	convergent
$ z $	6.1e-2	1.6e-2	5.4e-3	convergent
$ w $			oscillatory	conditionally convergent (it is the Fourier transform to k)

This row is different from the last three. There, z carried no phase and the large- $|z|$ end gave a genuine infrared logarithm. Here the $\mathfrak{C}$ delta leaves the factor $e^{-ik \cdot \frac{p^+}{k^+}(y'-z)}$, whose exponent grows with $|z|$, so the z integral is damped for every $p^+ \neq 0$ and converges on its own. **Nothing needs regulating.**

(For $|w|$: $\mathcal{P}^{ij}$ grows like $|w|$ through $\frac{(x-w)^i(x-z)^j}{2A}$, so the integrand only falls as $\frac{1}{|w|}$ and $|\langle F \rangle|$ as $R^{-3/2}$. That is not a divergence — the w integral is the Fourier transform to the measured momentum, and $\int d^2w \frac{e^{ik \cdot w}}{|w|} = \frac{2\pi}{|k|}$ converges conditionally. Do not bound it absolutely.)

5. The p^+ integral

Writing $D = A(p^+ + \beta)$ with $\beta = \frac{k^+ B}{A}$, and noting $A, B > 0$ so $\beta > 0$ and **there is no pole on the contour**, the p^+ dependence reduces to six masters. Because of the phase they are exponential integrals, not logarithms:

$$\mathcal{M}[h] = \int_{\Lambda}^{\vee - k^+} dp^+ \frac{h(p^+) e^{-i\kappa p^+}}{p^+ A + k^+ B}, \quad h \in \left\{ \frac{p^+}{S^2}, \frac{p^+}{S}, \frac{1}{S}, 1, \frac{1}{p^+}, p^+ \right\},$$

$$\int \frac{e^{-i\kappa p^+}}{p^+ + a} dp^+ = e^{i\kappa a} \text{Ei}(-i\kappa(p^+ + a)),$$

$$\mathcal{M}[1] = \frac{e^{i\kappa\beta}}{A} \text{Ei}(-i\kappa(p^+ + \beta)), \quad \mathcal{M}\left[\frac{1}{p^+}\right] = \frac{1}{A\beta} \left[\text{Ei}(-i\kappa p^+) - e^{i\kappa\beta} \text{Ei}(-i\kappa(p^+ + \beta)) \right],$$

$$\mathcal{M}[p^+] = \frac{1}{A} \left[\frac{e^{-i\kappa p^+}}{-i\kappa} - \beta e^{i\kappa\beta} \text{Ei}(-i\kappa(p^+ + \beta)) \right], \quad \mathcal{M}\left[\frac{1}{S}\right] = \frac{e^{i\kappa k^+} \text{Ei}(-i\kappa(p^+ + k^+)) - e^{i\kappa\beta} \text{Ei}(-i\kappa(p^+ + \beta))}{A(\beta - k^+)},$$

and $\mathcal{M}[\frac{p^+}{S}]$, $\mathcal{M}[\frac{p^+}{S^2}]$ follow from the same residues (the latter with a double pole at $p^+ = -k^+$). All six against quadrature at $A = 0.62$, $B = 1.45$, $\kappa = 0.83$, $\Lambda = 10^{-7}$, $\vee = 40$:

h=1	0.28052425-0.62934122j	agrees to 1.4e-15
h=1/p	10.74337225-0.80137719j	agrees to 1.8e-15
h=p	0.92920170+0.65250099j	agrees to 2.3e-14
h=1/S	0.32390943-0.33443150j	agrees to 4.0e-16
h=p/S	-0.04338518-0.29490972j	agrees to 1.4e-15
h=p/S^2	0.04623704-0.16604946j	agrees to 4.2e-16

Only $h = \frac{1}{p^+}$ is singular as $\Lambda \rightarrow 0$, and **what it produces is not** $\log \Lambda$:

$$\text{Ei}(-i\kappa\Lambda) \rightarrow \gamma_E + \log(|\kappa|\Lambda) - \frac{i\pi}{2} \text{sgn } \kappa = \gamma_E + \log \Lambda + \log \frac{|k \cdot (y' - z)|}{k^+} - \frac{i\pi}{2} \text{sgn } \kappa$$

(checked to six digits at $\kappa = 0.83, 2.4$ and $\Lambda = 10^{-6}, 10^{-8}$). **The rapidity logarithm arrives welded to a transverse logarithm.**

6. Can you use the + prescription? — No, not naively

At *fixed* transverse points there is nothing wrong with the identity. With $g(p^+) = \frac{e^{-i\kappa p^+}}{p^+ A + k^+ B}$ and $g(0) = \frac{1}{k^+ B}$,

$$\int_{\Lambda}^P \frac{dp^+}{p^+} g(p^+) = g(0) \log \frac{P}{\Lambda} + \int_0^P \frac{dp^+}{p^+} [g(p^+) - g(0)]$$

is exact. The damage is done by what happens next.

$g(0)$ deletes the z-phase. The only thing damping the z integral is $e^{-i\kappa p^+}$ with $\kappa = \frac{k \cdot (y' - z)}{k^+}$; setting $p^+ = 0$ removes it. Measured at large $ z $:			
$ z $	subtraction $g(0)$	full, $p^+ = 0.37$	full, $p^+ = 0.01$
10^2	0.212557193	1.26e-5	3.67e-3
10^3	0.212517864	3.30e-8	6.96e-6
10^4	0.212517471	1.12e-10	1.07e-8
10^5	0.212517467	5.48e-13	4.61e-11
The subtraction term sits on a flat plateau, equal to nine digits to			
$-\frac{1}{2} \frac{(x' - y') \cdot (x - w)}{(x' - y')^2 k^+ (x - w)^2} = -0.212517467,$			
so its z integral is logarithmically divergent . The unsubtracted term at any $p^+ \neq 0$ falls away. The prescription manufactures an infrared divergence that the row does not have.			

The mechanism is that the damping scale depends on p^+ :

$$|z|_{\max} \sim \frac{k^+}{|k| p^+} : \quad p^+ = 10^{-1} \rightarrow 12, \quad 10^{-2} \rightarrow 124, \quad 10^{-3} \rightarrow 1240, \quad 10^{-4} \rightarrow 12400,$$

visible in the table above as the $p^+ = 0.01$ column decaying later than the $p^+ = 0.37$ one. **The z and p^+ integrations do not commute.**

6.1 What to do instead

- There is no UV step.** Nothing to regulate before the p^+ integral.
- Do not split off $\log \frac{\sqrt{-k^+}}{\Lambda}$ at fixed z .** Either keep the z and p^+ integrals together, or introduce $d = 2 - 2\epsilon$ on z *first* — not because the row needs it, but because it is what makes the subtraction term finite and the split legitimate.
- Then the prescription is safe, but the logarithm is not the usual one.** By §5 it comes as $\log(|\kappa|\Lambda)$, so the $\log \frac{1}{\Lambda}$ multiplies a z integral that is itself log divergent, while its partner $\log |\kappa| \sim \log |z|$ against $\frac{d^2 z}{|z|^2}$ supplies a second logarithm. The row therefore produces a **mixed** $\frac{1}{\epsilon} \log \frac{1}{\Lambda}$ **and a** $\log^2$, not a clean single rapidity logarithm riding on a leading-order structure. Plan the absorption into evolution accordingly.

zint/src/row14_check.py (§1–2), row14_div2.py (§3–4, §6), row14_pplus.py, row14_plus.py (§5–6), row14_w.py (§4) — every number above.